

1. Do the following in sequence and record the results in a single program

Create a list with 5 different types of fruits. Print the list.

Add two more fruits to the list, then remove one fruit from it. Print the updated list.

Access the second and fourth fruits in the list. Print them.

Slice the list to get the first three fruits and print the result.

Find and print the length of your list.

```
# Create list of 5 fruits
```

```
fruits = ["apple", "banana", "mango", "orange", "grapes"]
```

```
print("Original list:", fruits)
```

```
# Add two, remove one
```

```
fruits.append("pineapple")
```

```
fruits.append("watermelon")
```

```
fruits.remove("banana")
```

```
print("Updated list:", fruits)
```

```
# Access second and fourth
```

```
print("Second fruit:", fruits[1])
```

```
print("Fourth fruit:", fruits[3])
```

```
# Slice first three
```

```
first_three = fruits[0:3]
```

```
print("First three fruits:", first_three)
```

```
# Find length
```

```
print("Length of list:", len(fruits))
```

Output:

Original list: ['apple', 'banana', 'mango', 'orange', 'grapes']

Updated list: ['apple', 'mango', 'orange', 'grapes', 'pineapple', 'watermelon']

Second fruit: mango

Fourth fruit: grapes

First three fruits: ['apple', 'mango', 'orange']

Length of list: 6

2. Do the following in sequence and record the results in a single program

Create a tuple with the names of 3 different cities you'd like to visit. Print the tuple.:

Access and print the first and last elements of the tuple.:

Create another tuple with 2 more cities. Concatenate both tuples and print the result.

Try changing one element of the tuple. What happens?

Unpack the elements of the tuple into separate variables and print them.

```
# Create tuple of 3 cities
```

```
cities = ("Delhi", "Mumbai", "Chennai")
```

```
print("Cities tuple:", cities)
```

```
# Access first and last
```

```
print("First city:", cities[0])
```

```
print("Last city:", cities[-1])
```

```
# Create another tuple and concatenate
```

```
more_cities = ("Kolkata", "Bengaluru")
```

```
all_cities = cities + more_cities
```

```
print("All cities:", all_cities)
```

```
# Try changing one element
```

```
# Unpack elements
```

```
c1, c2, c3 = cities
print("Unpacked cities:", c1, c2, c3)
```

Output:

Cities tuple: ('Delhi', 'Mumbai', 'Chennai')

First city: Delhi

Last city: Chennai

All cities: ('Delhi', 'Mumbai', 'Chennai', 'Kolkata', 'Bengaluru')

Unpacked cities: Delhi Mumbai Chennai

3. Do the following in sequence and record the results in a single program

Create a set of 5 unique colors. Print the set.

Add a new color to the set, then try removing an existing color. Print the updated set.

Create another set with 3 different colors. Find the intersection, union, and difference between both sets.

Check if a specific color is in the set and print the result.

Convert a list of fruits (with some duplicates) into a set and print the unique fruits.

```
# Create a set of 5 colors
colors = {"red", "blue", "green", "yellow", "black"}
print("Original set:", colors)

# Add one, remove one
colors.add("white")
colors.remove("blue")
print("Updated set:", colors)

# Set operations
other_colors = {"green", "purple", "orange"}
print("Intersection:", colors.intersection(other_colors))
```

```
print("Union:", colors.union(other_colors))
print("Difference:", colors.difference(other_colors))

# Check if color is in set
print("Is red in colors?", "red" in colors)

# List with duplicates to set
fruit_list = ["apple", "apple", "banana", "kiwi", "banana"]
unique_fruits = set(fruit_list)
print("Unique fruits:", unique_fruits)
```

Output:

```
Original set: {'blue', 'yellow', 'black', 'red', 'green'}
Updated set: {'white', 'yellow', 'black', 'red', 'green'}
Intersection: {'green'}
Union: {'yellow', 'purple', 'white', 'orange', 'red', 'black', 'green'}
Difference: {'red', 'black', 'yellow', 'white'}
Is red in colors? True
Unique fruits: {'kiwi', 'banana', 'apple'}
```

4. Do the following in sequence and record the results in a single program

Create a dictionary with your name, age, and favorite hobby as keys, and provide appropriate values. Print the dictionary.

Access and print the value associated with your name in the dictionary.

Add a new key-value pair for your favorite food, then update the value for your favorite hobby. Print the updated dictionary.

Print all the keys and all the values in the dictionary separately.

Remove the age key-value pair from the dictionary. Print the dictionary to see the change.

```
# Create dictionary
my_info = {"name": "Amman", "age": 23, "hobby": "gaming"}
print("Dictionary:", my_info)
```

```
# Access value of name
print("Name is:", my_info["name"])
```

```
# Add food, update hobby
my_info["food"] = "paneer butter masala"
my_info["hobby"] = "coding"
print("Updated dictionary:", my_info)
```

```
# Print keys and values separately
print("Keys:", my_info.keys())
print("Values:", my_info.values())
```

```
# Remove age
del my_info["age"]
print("After removing age:", my_info)
```

Output:

Dictionary: {'name': 'Amman', 'age': 23, 'hobby': 'gaming'}

Name is: Amman

Updated dictionary: {'name': 'Amman', 'age': 23, 'hobby': 'coding', 'food': 'paneer butter masala'}

Keys: dict_keys(['name', 'age', 'hobby', 'food'])

Values: dict_values(['Amman', 23, 'coding', 'paneer butter masala'])

After removing age: {'name': 'Amman', 'hobby': 'coding', 'food': 'paneer butter masala'}